座位号________ 专业________ 学院________ 学号□□□□□□□□□□□□ 姓名________

（密封线内不答题）

密 封 线

诚信应考,考试作弊将带来严重后果!

华南理工大学期末考试

2023 Calculus Mid-Term Exam

注意事项：1. 考前请将密封线内填写清楚；
2. 所有答案请直接答在试卷上；
3. 考试形式：闭卷；
4. 本试卷共 9 大题，满分 100 分， 考试时间 120 分钟。

题 号	1，2	3， 4	5， 6	7， 8	9	总分
得 分						
评卷人						

1. Answer the questions (30):

(1) Does the series $\sum_{n=1}^{\infty}\left(\frac{\sin na}{n^2}-\frac{1}{\sqrt{n}}\right)$ converge or diverge?

(2) Find the volume of the parallelepiped with edges $3\vec{i}-4\vec{j}+2\vec{k}$, $-\vec{i}+2\vec{j}+\vec{k}$ and $3\vec{i}-2\vec{j}+5\vec{k}$

(3) Suppose $x^3e^{y+z}-y\cos(2xz)=0$, find $\partial z/\partial x$

(4) Let $z=f\left(2x\ ,\frac{y^2}{x}\right)$ has the second-order continuous partial derivatives, find $\frac{\partial^2 z}{\partial x\partial y}$.

(5) Find the equation of the plane through $(-1,-2,3)$ and perpendicular to both the plane $x-3y+2z=7$ and $2x-2y-z=-3$

(6) Find ∇f and $D_u f$, where $f=x^3+\tan yz$, $u=(2,0,-4)$

2. Evaluate the problems (30):

(1) Rewrite the iterated integral with the indicated order of integration

$\int_{-1}^{0}\int_{-\sqrt{y+1}}^{\sqrt{y+1}} f(x,y)dxdy, \qquad dydx$

(2) Switch to rectangular coordinate and evaluate $\int\limits_{3\pi/4}^{4\pi/3} d\theta \int\limits_{0}^{-5\sec\theta} r^3\sin\theta dr$

(3) Switch the order and then evaluate the integrals:

$\int_{\frac{1}{4}}^{\frac{1}{2}}\mathrm{d}y\int_{\frac{1}{2}}^{\sqrt{y}} e^{\frac{y}{x}}\mathrm{d}x+\int_{\frac{1}{2}}^{1}\mathrm{d}y\int_{y}^{\sqrt{y}} e^{\frac{y}{x}}\mathrm{d}x$

(4) Evaluate $\int\limits_{0}^{1}\int\limits_{0}^{\sqrt{1-x^2}} \frac{1}{\sqrt{4-x^2-y^2}}dydx$

(5) Find $\iint\limits_{D}\frac{y}{x+y}e^{(x+y)^2}\mathrm{d}x\mathrm{d}y, D$ is bounded by $x+y=1, x=0$ and $y=0$.

(6) Compute the triple integral $\iiint\limits_{\Omega} zdv$, where $\Omega:\begin{cases}x^2+y^2+z^2\le 2\\ z\ge x^2+y^2\end{cases}$

3 (5) Find the minimum and maximum for the function $z^2 = x^2 + y^2$ with the constraint $(x-\sqrt{2})^2 + (y-\sqrt{2})^2 \le 9$.

4 (5) Find convergence set and sum for the power series $\sum_{n=1}^{\infty} n(n+1)(x-1)^n$

5 (5) If $f(x,y) = \begin{cases} \dfrac{x^3 y}{x^2+y^2}, x^2+y^2 \neq 0 \\ 0, \qquad\quad x^2+y^2 = 0 \end{cases}$, find $f_{xy}(0,0)$.

6 (5) Calculate $\int_0^3 \int_0^{\sqrt{9-x^2}} \int_0^2 (x^2+y^2)^{1/2} dzdydx$

7 (5) Find the parametric equation of the line that is tangent to the curve of intersection of the surface $\mathrm{x} = \mathrm{z}^2$ *and* $y = z^3$ at $(1, 1, 1)$

8 (5) Find the volume of the solid bounded by the surface $z = xy$ and planes $z = x + y$, $x + y = 1$, $x = 0, y = 0$.

9. (5) Consider the Cobb-Douglas production model for a manufacturing process depending on three input x, y, z with unit costs a, b, c , respectively. Given by

$$P(x,y,z) = k\,x^{\alpha} y^{\beta} z^{\gamma} \quad \alpha > 0, \beta > 0, \gamma > 0, \quad \alpha + \beta + \gamma = 1$$

subject to the cost constrain $ax + by + cz = 1$. Determine the x, y, z maximize the production

.

10. (5) Evaluate $\displaystyle\int_{-3}^{3} \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} \int_{-\sqrt{9-x^2-z^2}}^{\sqrt{9-x^2-z^2}} (x^2 + y^2 + z^2)^{3/2} dydzdx$